Ku-Yu Fan

PhD Candidate in Mathematics | Nagoya University

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RESEARCH INTERESTS

Multiple zeta values, motivic periods and coactions, Schur and arborified multiple zeta values, p-adic multiple zeta values, multiple polylogarithms and p-adic multiple L-functions.

EDUCATION

2024 - 2027 (expected) **Doctor of Philosophy (Mathematics), Graduate School of Mathematics, Nagoya University (NU) (Advisor: Hidekazu Furusho) [PhD expected September 2027]**

2021-2024 **Master of Science (Mathematics), Institute of Mathematics, National Taiwan University (NTU) (Advisor: Nobuo Sato)**

2017-2021 **Bachelor of Science (Mathematics), Department of Mathematics, National Tsing Hua University (NTHU)**

REFEREED PUBLICATIONS

2026 **A map between arborifications of multiple zeta values**

Ku-Yu Fan | Journal of Algebra | arXiv:2508.20387

2026 **Coaction formula for the motivic version of Yamamoto's integral**

Ku-Yu Fan | Mathematische Zeitschrift | Vol. 314, Article 35 | DOI: 10.1007/s00209-026-04138-w | arXiv:2406.08127

2021 **The number of lattice lines in lattice figures [in Chinese]**

Ku-Yu Fan, Feng-Mei Lin | Mathematical Media [in Chinese] | Vol. 45, No. 2, pp. 44-53

PREPRINTS

2026 **Regularised double shuffle relations for planar Arborified Zeta Values**

Pierre Catoire, Pierre J. Clavier, Ku-Yu Fan | arXiv:2609.02346 [math.NT]

2026 **p -adic multiple L -functions and twisted multiple Bernoulli numbers**

Ku-Yu Fan | arXiv:2607.04086 [math.NT]

2026 **p -adic multiple zeta values of integer indices**

Ku-Yu Fan | arXiv:2603.22923 [math.NT]

SELECTED RESEARCH TALKS

2026-09 **Explicit Formulas for Cyclotomic Multiple Bernoulli Numbers**

NCTS Number Theory Seminar | National Tsing Hua University

2026-09 **An explicit formula for the special values of the p-adic multiple L-functions**

MSJ Autumn Meeting 2026 | Kobe University

2026-07 **Double shuffle relations for p-adic multiple zeta values of integer indices**

The 5th International Workshop on Multiple Zeta Values and Related Topics | Central South University

2026-07
Double shuffle relations for integer indices and their application to -adic multiple zeta values
25th Sendai-Hiroshima Number Theory Conference | Tohoku University

2026-06
p-adic multiple zeta values and double shuffle relations for integer indices
Algebraic and Combinatorial Perspectives in the Mathematical Sciences | Online

2026-03
p-adic multiple zeta values of integer indices
Meeting in the Middle: Conference on characteristic 0 and characteristic p multiple zeta values (and other topics) | University of Hawai'i at Manoa

2026-02
p-adic multiple L-function and twisted multiple Bernoulli numbers
MATHSCI FRESHMAN SEMINAR 2026 | Nagoya University

2025-09
A Map Between Arborifications of Multiple Zeta Values
2025 RIMS Symposia (open) "Various Aspects of Multiple Zeta Values" | Kyoto University

2025-09
A Map Between Arborifications of Multiple Zeta Values
MSJ Autumn Meeting 2025 | Nagoya University

2025-07
A Map Between Arborifications of Multiple Zeta Values
The 4th International Workshop on Multiple Zeta Values and Related Topics | Tongji University

2024-11
Coaction Formula for the Motivic Version of Yamamoto's Integral
2nd Kindai Workshop Multiple Zeta Values and Modular Forms | Kindai University

2024-05
Coaction formula for the motivic version of Yamamoto's integral
Monthly Meeting | Online

AWARDS AND FELLOWSHIPS

2024-2027
Japan-Taiwan Exchange Association Scholarship (Graduate Research Students)
Research fellowship / scholarship

2019
Professor Han-Ching Lai Memorial Scholarship
Research fellowship / scholarship

TEACHING EXPERIENCE

Spring 2026
Special Topics in Foundations (Special Topics in Foundations)
Teaching Assistant, Nagoya University

Fall 2025
Calculus
Teaching Assistant, Nagoya University

Spring 2025
Calculus
Teaching Assistant, Nagoya University

Spring 2024
Calculus
Teaching Assistant, National Taiwan University

Fall 2023
Calculus
Teaching Assistant, National Taiwan University

Spring 2023
Calculus
Teaching Assistant, National Taiwan University

Spring 2023
Modern Algebra
Teaching Assistant, National Taiwan University

Fall 2022
Calculus
Teaching Assistant, National Taiwan University

Fall 2022
Modern Algebra
Teaching Assistant, National Taiwan University

Spring 2022
Linear Algebra
Teaching Assistant, National Taiwan University

Fall 2021
Linear Algebra
Teaching Assistant, National Taiwan University

Spring 2021
Elementary Number Theory
Teaching Assistant, National Tsing Hua University

Spring 2021
Calculus
Teaching Assistant, National Tsing Hua University

Fall 2020
Calculus
Teaching Assistant, National Tsing Hua University

Spring 2020
Advanced Calculus
Teaching Assistant, National Tsing Hua University

ACADEMIC SERVICE AND ORGANIZATION

2026
Nagoya-Kansai Young Researchers' Zeta Meeting - Organizing Committee (2026; joint 19th Zeta Young Researchers' Meeting / 70th Kansai MZV Seminar; supported by a selected 2025 Nagoya University student project)
Organizing role; see website for details

2024-
Monthly Research Seminar (MZV) - Co-organizer (2024-present)
Organizing role; see website for details

SELECTED PATENTS

2017-2021
Geometric puzzles and mathematics education
Three Taiwan utility models (M548578, M566100, M612210) and two US patents (10,322,337; 10,722,782); joint inventor with Wen-Liang Hung.